DBMS

Data → facts and details about something or someone

Information → the processed form which is meaningful and useful for decision making.

RDBMS → Relational database ~~management~~ system

→ Collection of data which are interrelated to each other and stored in the form of table. (2) relation

SQL

DBMS → Database Management System

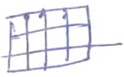
The software that helps us to create, maintain, manage database

~~eg~~ eg MySQL, Oracle, etc

MySQL is a software used to create some database using language like SQL

Instance

at particular point of time what data currently exist in a db



Schema

overall logical structure, how Table is organised, no of rows, no of columns, etc.

OLAP → Online Analytical processing

OLTP → Online Transaction processing

OLAP → historical data used for decesion making.

eg → old deleted accounts in SBI (85+)

OLTP → daily active transaction will be handeld, very quickly, and fastly.

→ present account working acc in (BI

DBA
Database Administrator
The person responsible for Managing securing, maintaing database to ensure they Run smoothly and efficiently

Data Warehousing
collect data and Gathering it from diff from diff source, and storing all data together in a single place to Analyse it efficiently, to make the decision effectively.
eg → storing, student marks, attendence student feebacks in single place to Analyse overall student perfomence.

Data Mining
process of discovering hidden patterns R/p and usefull info from the data we hv stored to make effective decision and predict future trends.

Data science
similrly extracting meaningfull insights from data using scientific Method algorithms and techniques to solve real-World problem.

Data Warehouse → stores data
Data Mining → finds hidden patterns.
Data science → Combines all these to Predict And solve problems.

ER- Diagram

Graphical diagram of database

 → Tables in a database

 → show r/p b/w ~~data~~ tables in db

 → Attributes of table @ Col

Entity

In a Table every Row is an entity

Entity set

Table is entity set

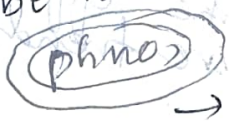
Attributes are the cols, that describes the properties of each member in entityset (Table)

Types

→ Single attribute vs Multiple,

Single → Only have one value →  

multiple → maybe have multiple value

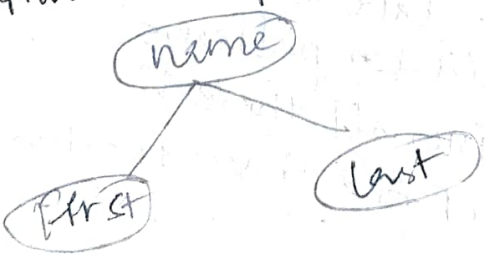


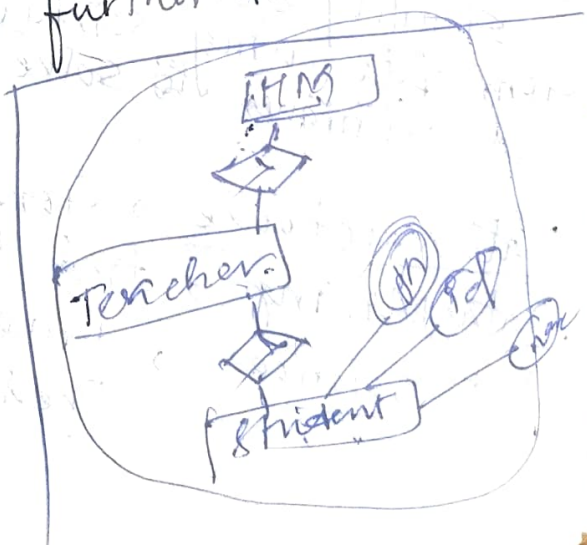
→ double ellipse

→ Simple attributes vs composite

attribute cannot be divided further

Composite → attributes further divided into sub parts





→ stored attribute vs derived

→ will store value permanently stored in db.
eg → dob (dob)

derived → derived from other Attributes
age from dob

(dob) (age) → dotted ellipse

Relation ships

1) Unary →
2) Binary →
3) Ternary →
4) N-ary relationship →

one → One → person to one adhar
one → n → person to multiple phone no
n → One → ph no to person
n → n → doctor to patients

Strong entity set → has primary key in table
Weak entity set → dont have primary key in table

→ Rows / Tuple / cardinality / Record → row
→ Columns / Field / column / Arity / degree → Cols
Arity / degree → no of Cols.

#

Properties of Relational Table

1) each cell Contain atomic (unique, single) Value
2) Val in Cols are same kind,
   eg → In city Col → only cities
3) Each row is unique
4) No 2 table have same schema
   same name
5) each col has unique name in table
6) sequence of rows & cols are not matter

Normalisation

When we too much info in a single table it becomes problem to insert, delete & to Modify, so, to solve this, we devide huge table into sub table to overcome above problems.

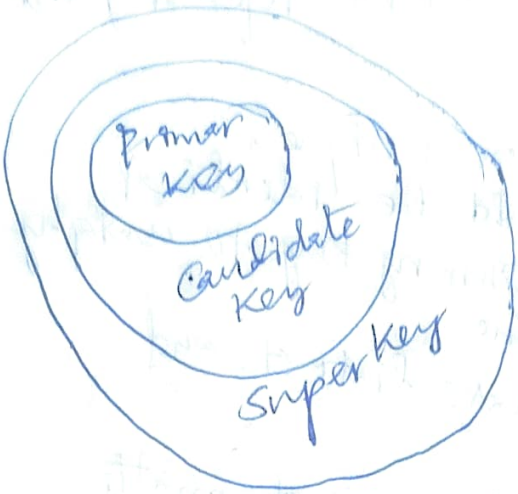
Functional dependencies

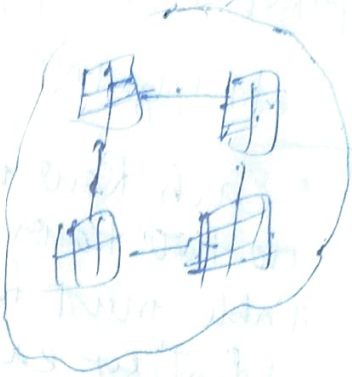
$\alpha \rightarrow \beta$

using unique given no we have unique identify unique related value.

Key

for identifying the uniqueness of a tuple





Superkey → One or More Combin' of cols. are there to uniquely identify every row in the table.

candidate key : minimal super key. small set of cols can uniquely identify every Row., if we remove any col from it, it loses its uniqueness.

Primary key → specific candidate key ~~that~~ key

→ Value → Only one primary key for table..

Composite → (It can be super, candidate, primary).

Normalisation is a structured process used in database design to Minimise Redundancy and to avoid data inconsistency It involve breaking down larger table into smaller, table while preserving all essential info and R/ps.

- Remove duplicates
- By dividing table into sub tables,

First Normal form → Table defaultly 1NF

To satisfy 1NF

- Each Row must be unique,
- no two Rows should be identical
- Table must have primary key to uniquely identify each Row
- Each Col should have distinct and meaningful name
- Order of Rows and Cols does not matter in Rln table

Candidate Key



R (A B C D)

A B → C D

A B → Candidate key

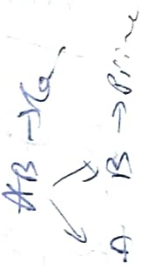


Prime attribute

attributes is part of any of candidate keys
Cols
here A & B

Non-prime attribute

Attribute if it is not part of any candidate key here C and D

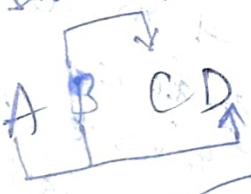


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partial dependency

When a non prime attribute is dependent only on a part (subset) of candidate key,

eg → B → C



part of prime → non-prime

Total dependency

When a non-prime attribute dependent on the entire candidate key

eg → AB → D

2NF

- R should be in 1NF
- R should not contain any partial dependency (any non prime attribute should be fully dependent upon candidate key)

Transitive dependency

R(A B C)

A is candidate key (prime attribute)

A → B

B → C → here non-prime attribute

B identifying non-prime attribute of C

dependency from non-prime attribute to non prime attribute is transitive dependency

B → C

3NF

- R should be in 2NF
- not contain any transitive dependency.

BCNF

Boyce codd Normal form

- should not contain partial dependency and trasitive depency no-non-prime attribute

If there is any dependency from $\alpha \rightarrow \beta$ then that α must be a super key.

Transaction

unit of work in the db by some operations (read, write, update) either all has to sucessfully has to be done or fail

eg @ ntg → to → amount debit from @ credit

properties of transition

ACID

A → atomicity
C → consistency
I → Isolation
D → Durability.

atomicity → transation is an atomic unit of processing, it should either be performed in its entirely @ not performed at all.

Consistency

The DB should Remain Consistent before and after transaction

Isolation

Execution of one Transaction should not be interfered with any other transaction executing Concurrently

Durability

Changes applied to db by transaction must persist in db

a = ~~10~~ 15

a from 10 to 15 → changes has to persist not lost